BTS-VII(R/S)-11-23-1977 Reg. No. [ ] **A**

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2023

ME 19-205-0706 AUTOMOBILE ENGINEERING
(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Compare the working of various types of automobiles.
CO2: Learn the functions and working of different engine components.
CO3: Appreciate the use of alternate fuels.
CO4: Understand the latest technology used in the field of automobile.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Discuss combustion chamber types.	3	L2	2	
	(b)	What do you mean by 'valve clearance'?	3	L2	3	
	(c)	Name the main components of fuel supply system for automotive diesel engines with the help of a neat diagram	3	L2	2	
	(d)	What are the advantages of using CNG as an alternate fuel?	3	L2	3	
	(e)	What are the functions of transmission in a vehicle?	3	L2	2	
	(f)	What are the advantages of constant mesh gear box when compared with sliding mesh type gear box?	3	L2	2	
	(g)	Mention the advantages of hybrid vehicles.	3	L2	4	
	(h)	Draw the basic block diagram of electric vehicle showing its components and explain.	3	L2	4	

PART B

		(4 × 12 = 48)				
II.	(a)	Define the term Piston slap. What is the cause of piston slap?	4	L2	1	
	(b)	Discuss in detail about the different materials used to manufacture pistons, cylinders, connecting rods and crankshaft.	8	L2	2	
		OR				
III.	(a)	Explain VTEC technology to effect variable valve timing in automobiles with a neat sketch.	6	L2	4	
	(b)	Discuss ignition advance and its features.	6	L2	2	

(P.T.O.)

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		Question	Marks	BL	CO	PO
IV.	(a)	What are the different types of fuel supply systems for a petrol engine?	2	L2	2	
	(b)	What is the necessity of a cooling system? Explain a cooling system with a neat sketch.	10	L3	2	
		OR				
V.	(a)	What are the requirements of an ignition system of an internal combustion engine?	2	L3	2	
	(b)	Discuss in detail about the battery ignition system. Illustrate your answer with neat sketches.	10	L2	2	
VI.	(a)	Explain drive train of an automobile.	2	L2	1	
	(b)	Explain the necessity of a differential in an automobile. Discuss in detail the operation of a differential with a neat sketch.	10	L2	2	
		OR				
VII.	(a)	What is a CVT? Discuss its main advantages and limitations.	5	L2	4	
	(b)	Explain the working of synchromesh gear box with neat sketches.	7	L2	2	
VIII.	(a)	Explain the concept of electric traction.	6	L2	4	
	(b)	Discuss in detail about Electric Vehicle Drive train alternatives based on Power source configuration.	6	L2	4	
		OR				
IX.	(a)	Discuss in detail about energy storage systems in hybrid electric vehicles.	8	L2	4	
	(b)	What do you mean by high energy density and high power density? Explain with examples.	4	L2	4	

Bloom's Taxonomy Levels
L2 = 90%, L3 = 10%.

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